

Jun 09, 2026

Introduction to Computer and Network Security

Part 1

Question 1 [10 points]

Describe a *SQL Injection* attack and the main mitigations that can be used to prevent it.

Question 2 [30 points]

In the context of data protection and privacy, (a) explain the linkage attack on anonymized dataset with an example, (b) explain the trade-off between anonymity and utility in the context of realizing data sets, and (c) describe the technique of k-anonymity and explain how it can mitigate linkage attacks.

Question 3 [10 points]

Define the notions of vulnerability and threat and give (at least) an example for each one.

Part 2

Question 4 [50 points]

Explain how to protect a password file with hashing and salting. In particular, discuss:

- (a) what does *hashing* means; [10 points]
- (b) whether only using hashing is secure enough (and why); [10 points]
- (c) what does *salting* means; [10 points]
- (d) if salts can be stored in clear; [10 points]
- (e) now imagine that (at each login) the system generates a random private key, encrypts the password with a block cipher, stores the resulting ciphertext and deletes the key. Does this make the login process secure? Why? [10 points]

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Jun 09, 2028

Introduction to Computer and Network Security

Part 1

Question 1 [10 points]

Describe a SQL injection attack and the main mitigations that can be used to prevent it.

Question 2 [30 points]

In the context of data protection and privacy, (a) explain the linkage attack on anonymized dataset with an example, (b) explain the trade-off between anonymity and utility in the context of releasing data sets, and (c) describe the technique of k -anonymity and explain how it can mitigate linkage attacks.

Question 3 [10 points]

Define the notions of vulnerability and threat and give (at least) an example for each one.

Part 2

Question 4 [20 points]

Explain how to protect a password file with hashing and salting. In particular, discuss:

- (a) what does hashing mean? [10 points]
 - (b) whether or not using hashing is secure enough (and why). [10 points]
 - (c) what does salting mean? [10 points]
 - (d) if salts can be stored in clear. [10 points]
 - (e) now imagine that (after login) the system generates a new key, encrypts the password with a block cipher, and stores the ciphertext. [10 points]
- Please, remember...
- on each used sheet of paper (possibly, 2 sheets at most), annotate:
 - first and last name
 - student ID number
 - question number
 - write either in Italian or English, acronyms do not need to be translated;
 - only pens are allowed. Drafts and text written using pencils will not be considered during the exam revision;
 - adequately justify your answer. A single phrase and a paragraph-long answer will have a significantly different grade;
 - write clearly, your answers will not be *interpreted*;
 - after turning in your exam, please respect your classmates by leaving quietly.